

EXPERIMENT NO: - 06

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Class:- D15A

Roll:No:- 24

AIM:- To connect Flutter UI with Firebase database.

Theory: -

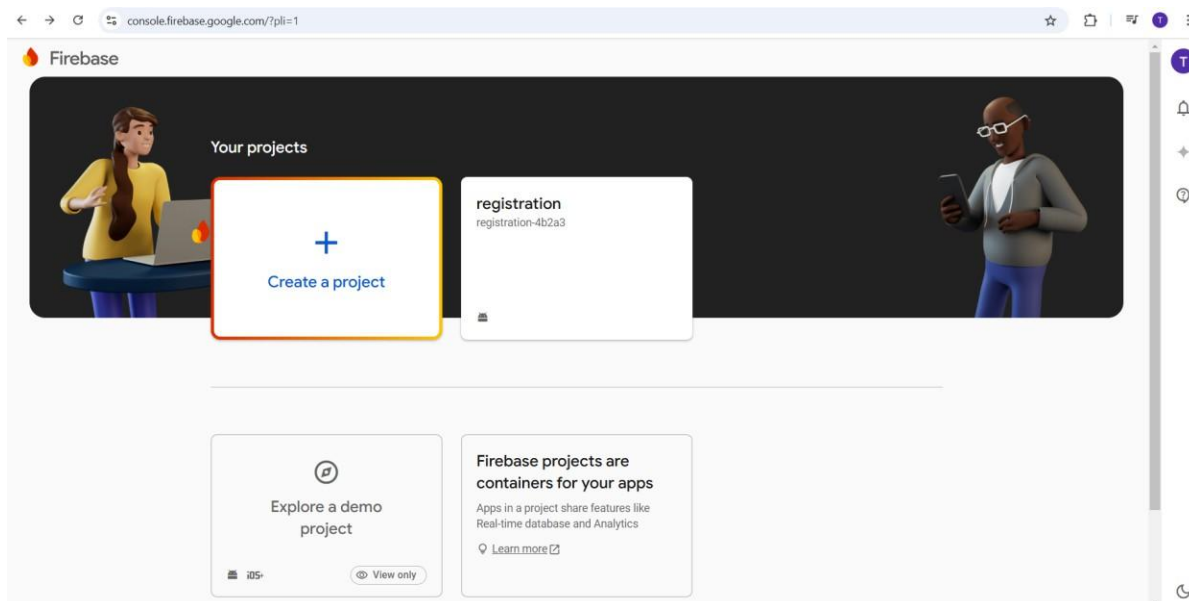
Flutter is an open-source UI toolkit developed by Google for building natively compiled applications for mobile, web, and desktop from a single codebase. Firebase, a Backend-as-a-Service (BaaS) platform, provides real-time database, authentication, and cloud storage services, making it a powerful backend solution for Flutter applications.

By integrating Firebase with Flutter, developers can store and retrieve data in real time, authenticate users, and manage cloud-based data efficiently. This is particularly useful for applications requiring dynamic content updates and user interactions.

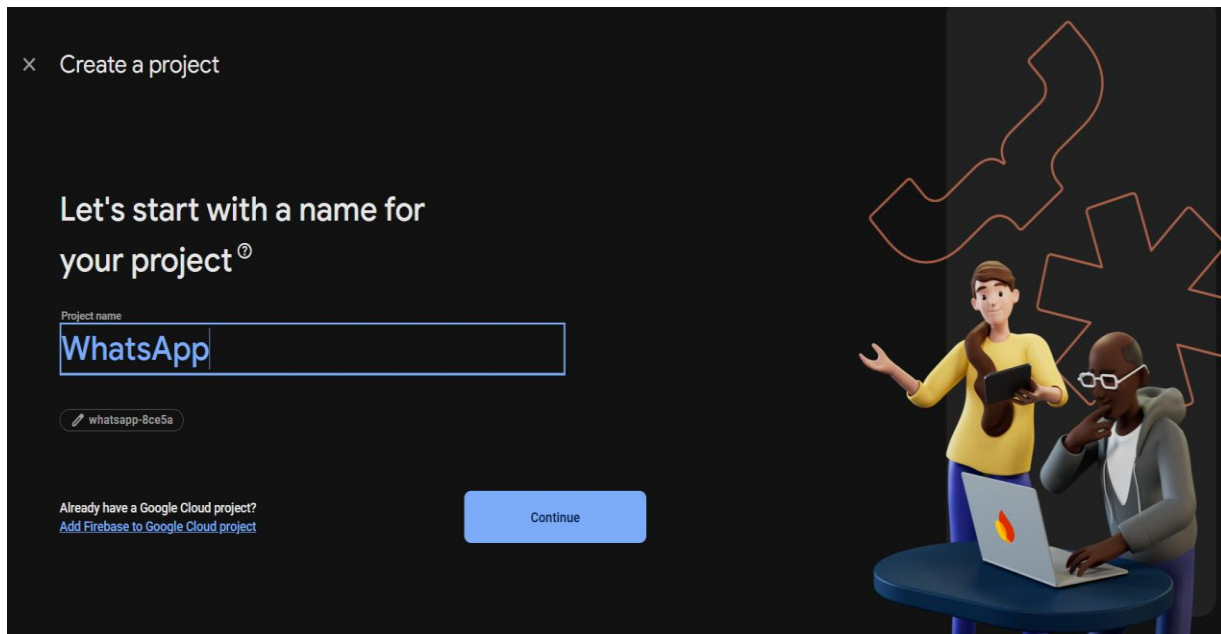
➤ **Steps to Connect Flutter UI with Firebase Database**

Step 1:

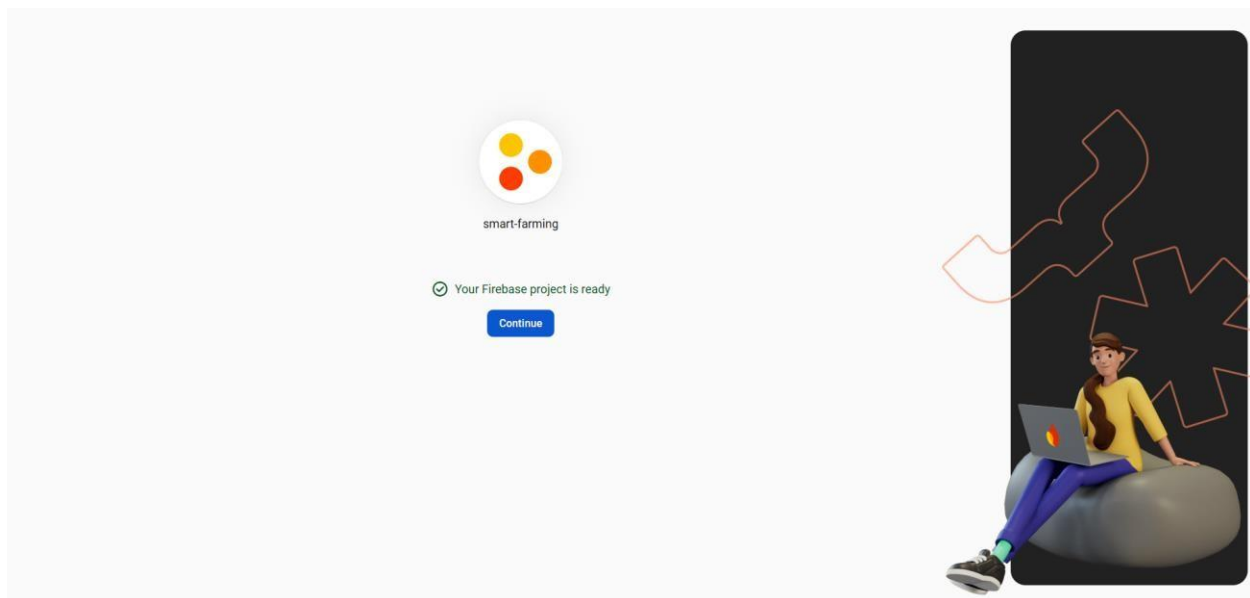
1.1) Go to Firebase Console and Create a Firebase Project



1.2) Click on Create a Project and give it a suitable name.



1.3) Enable Google Analytics (optional) & Click continue and complete the setup



2.1) Click on Android/iOS/Web based on your Flutter application

smart-farming

Project Overview

Generative AI

Build with Gemini

Genkit

Product categories

Build

Run

Analytics

All products

Related development tools

IDX

Checks

Spark

Upgrade

smart-farming

Receive email updates about new Firebase features, research and events

Sign up

smart-farming

Get started by adding Firebase to your app

Android

iOS

Android

Web

Flutter

Get started by adding Firebase to your app

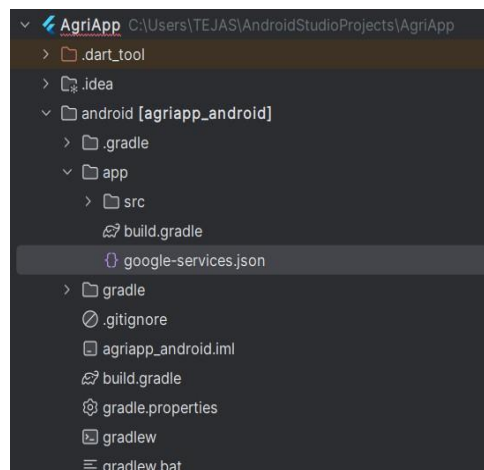
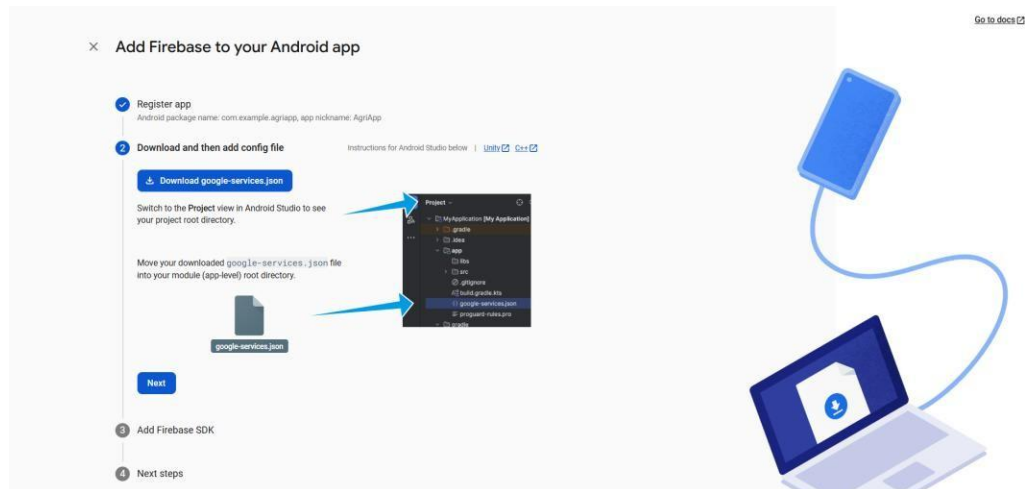
Experiment with a Gemini 2.0 sample app

Learn about the Gemini API and how it works by experimenting with an AI-powered sample app

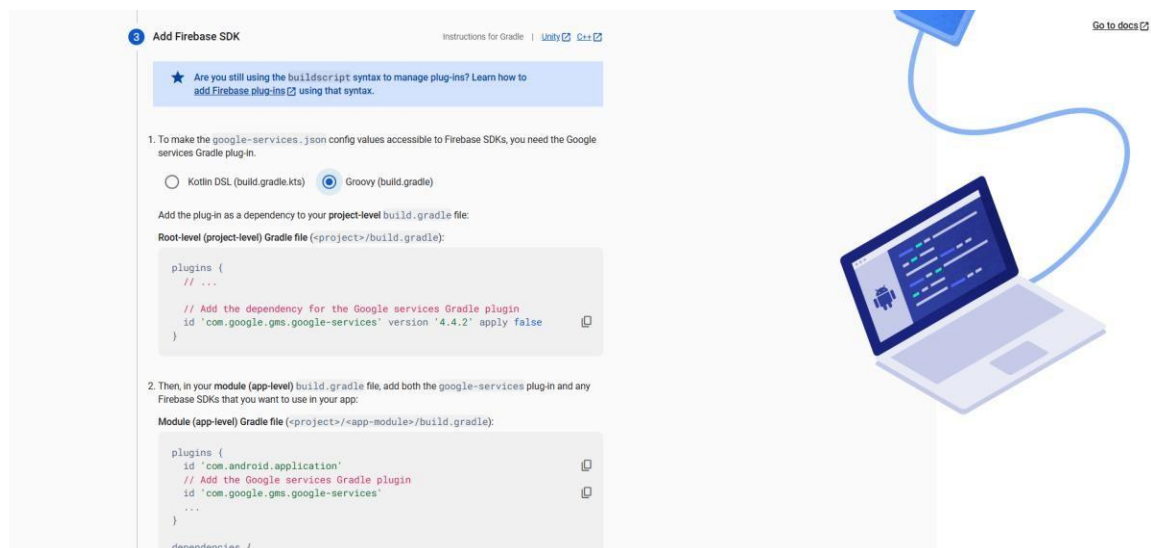
Try it now

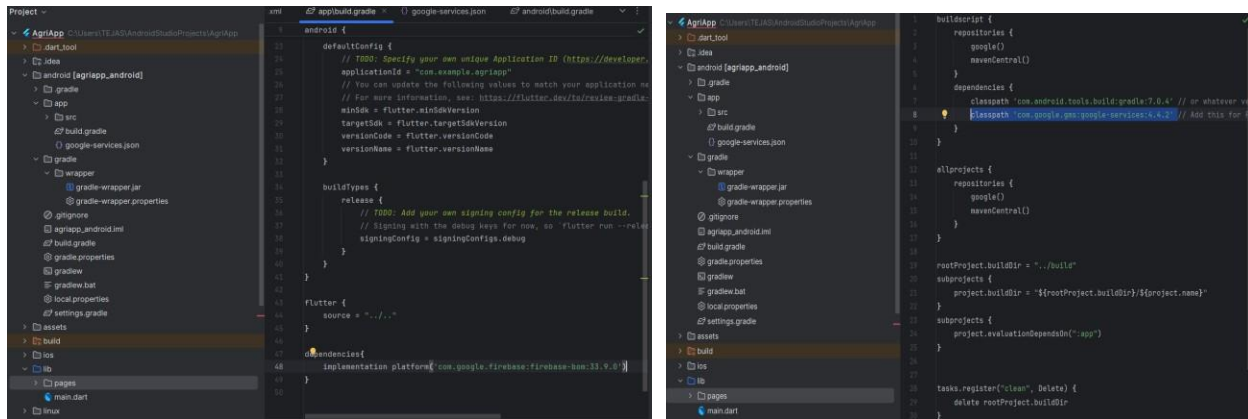
[illegible]

2.3) Download the google-services.json (for Android) & place the JSON file inside android/app/ directory.



2.4) Add Firebase SDK dependencies to android/build.gradle





Step 3: - Add Firebase Authentication to Your App

3.1) Add Firebase Authentication Dependencies

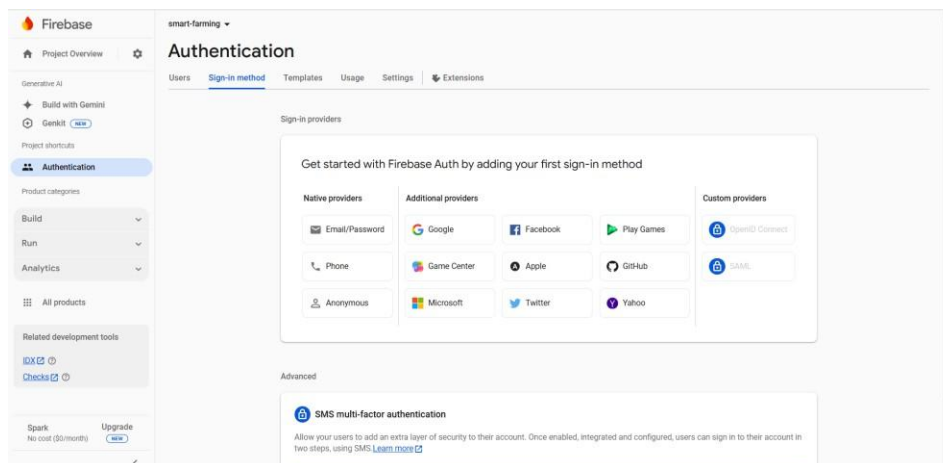
```
dependencies:
  flutter:
    sdk: flutter
  firebase_core: ^3.11.0
  firebase_auth: ^5.4.2 # For authentication
  cloud_firestore: ^5.6.3 # For Firestore, if you need it
  firebase_messaging: ^15.2.2
  http: ^0.13.3
  image_picker: ^1.0.4
  tf_lite_flutter: ^0.11.0
  image: ^3.2.0
  url_launcher: ^6.1.14
```

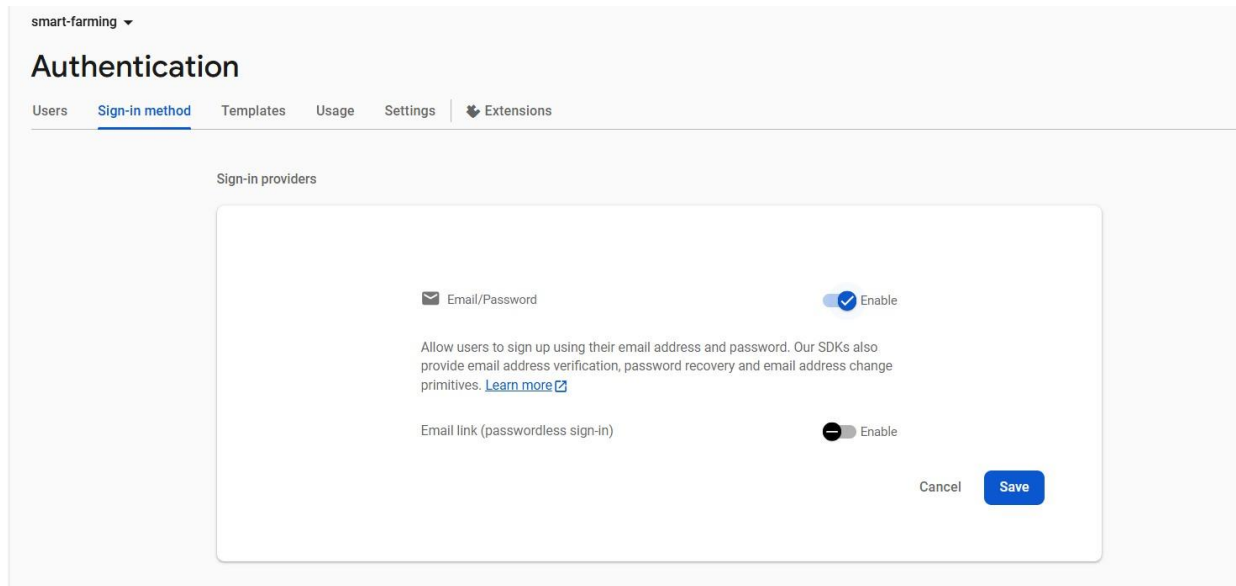
3.2) Enable Authentication in Firebase Console

Go to **Firebase Console** → **Authentication**.

Click on **Sign-in method** and enable **Email/Password** (or any other method like Google).

Click Save





3.3) Implement Authentication in Flutter

Modify main.dart

```
import 'package:firebase_core/firebase_core.dart';
import 'package:firebase_auth/firebase_auth.dart';

void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp();
  runApp(MyApp());
}
```

Step 4: -Configure Firebase Realtime Database

- 4.1) Go to Firebase Console → Realtime Database.
- 4.2) Click **Create Database** → Choose location → Set rules (for development, set read/write to true).
- 4.3) Click **Publish**.

Code:-

Login page:

```
import
'package:country_picker/country_picker.dart
';
import 'package:flutter/material.dart';
import
'package:flutter_riverpod/flutter_riverpod.da
rt';
import
'package:whatsapp_messenger/common/exte
nsion/custom_theme_extension.dart';
import
'package:whatsapp_messenger/common/hel
per/show_alert_dialog.dart';
import
'package:whatsapp_messenger/common/wid
gets/custom_elevated_button.dart';
import
'package:whatsapp_messenger/feature/auth/
controller/auth_controller.dart';
import
'package:whatsapp_messenger/feature/auth/
widgets/custom_text_field.dart';

import '../../common/utils/colours.dart';
import
'../../common/widgets/custom_icon_button
.dart';

class LoginPage extends
ConsumerStatefulWidget {
  const LoginPage({super.key});

  @override
  ConsumerState<LoginPage> createState()
=> _LoginPageState();
}

class _LoginPageState extends
ConsumerState<LoginPage> {
```

```
late TextEditingController
countryNameController;
late TextEditingController
countryCodeController;
late TextEditingController
phoneNumberController;

sendCodeToPhone() {
  final phoneNumber =
phoneNumberController.text;
  final countryName =
countryNameController.text;
  final countryCode =
countryCodeController.text;

  if (phoneNumber.isEmpty) {
    return showAlertDialog(
      context: context,
      message: "Please enter your phone
number",
    );
  } else if (phoneNumber.length < 9) {
    return showAlertDialog(
      context: context,
      message:
        "The phone number you entered is
too short for the country:
$countryName\n\nInclude your area code if
you haven't",
    );
  } else if (phoneNumber.length > 10) {
    return showAlertDialog(
      context: context,
      message:
        "The phone number you entered is
too long for the country: $countryName",
    );
  }
}
```

```

        ref.read(authControllerProvider).sendSms
        Code(
            context: context,
            phoneNumber:
            "+$countryCode$phoneNumber",
        );
    }

```

```

showCountryPickerBottomSheet() {
    showCountryPicker(
        context: context,
        showPhoneCode: true,
        favorite: ['ET'],
        countryListTheme:
        CountryListThemeData(
            bottomSheetHeight: 600,
            backgroundColor:
            Theme.of(context).colorScheme.surface,
            flagSize: 22,
            borderRadius:
            BorderRadius.circular(20),
            textStyle: TextStyle(color:
            context.theme.greyColor),
            inputDecoration: InputDecoration(
                labelStyle: TextStyle(color:
                context.theme.greyColor),
                prefixIcon: const Icon(
                    Icons.language,
                    color: Colors.greenDark,
                ),
                hintText: 'Search country by code or
                name',
                enabledBorder:
                UnderlineInputBorder(
                    borderSide: BorderSide(
                        color:
                        context.theme.greyColor!.withOpacity(.2),
                    ),
                ),
                focusedBorder: const
                UnderlineInputBorder(
                    borderSide: BorderSide(

```

```

            color: Colors.greenDark,
        ),
    ),
    ),
    ),
    onSelect: (country) {
        countryNameController.text =
        country.name;
        countryCodeController.text =
        country.phoneCode;
    },
);
}

@override
void initState() {
    countryNameController =
    TextEditingController(text: 'Ethiopia');
    countryCodeController =
    TextEditingController(text: '251');
    phoneNumberController =
    TextEditingController();
    super.initState();
}

@override
void dispose() {
    countryNameController.dispose();
    countryCodeController.dispose();
    phoneNumberController.dispose();
    super.dispose();
}

@override
Widget build(BuildContext context) {
    return Scaffold(
        appBar: AppBar(
            backgroundColor:
            Theme.of(context).scaffoldBackgroundColo
            r,
            elevation: 0,

```



```

        title: Text(
          'Enter your phone number',
          style: TextStyle(
            color:
context.theme.authAppBarTextColor,
          ),
        ),
        centerTitle: true,
        actions: [
          CustomIconButton(
            onPressed: () {},
            icon: Icons.more_vert,
          ),
        ],
      ),
      body: Column(
        children: [
          Padding(
            padding: const
EdgeInsets.symmetric(horizontal: 20,
vertical: 10),
            child: RichText(
              textAlign: TextAlign.center,
              text: TextSpan(
                text: 'WhatsApp will need to
verify your number. ',
                style: TextStyle(
                  color: context.theme.greyColor,
                  height: 1.5,
                ),
              children: [
                TextSpan(
                  text: "What's my number?",
                  style: TextStyle(
                    color:
context.theme.blueColor,
                  ),
                ),
              ],
            ),
          ),
        ],
      ),

```

```

        ),
      ),
      const SizedBox(height: 10),
      Padding(
        padding: const
EdgeInsets.symmetric(horizontal: 50),
        child: CustomTextField(
          onTap:
showCountryPickerBottomSheet,
          controller: countryNameController,
          readOnly: true,
          suffixIcon: const Icon(
            Icons.arrow_drop_down,
            color: Colors.greenDark,
            size: 22,
          ),
        ),
      ),
      const SizedBox(height: 10),
      Padding(
        padding: const
EdgeInsets.symmetric(horizontal: 50),
        child: Row(
          children: [
            SizedBox(
              width: 70,
              child: CustomTextField(
                onTap:
showCountryPickerBottomSheet,
                controller:
countryCodeController,
                prefixText: '+',
                readOnly: true,
              ),
            ),
            const SizedBox(width: 10),
            Expanded(
              child: CustomTextField(
                controller:
phoneNumberController,

```

```

        hintText: 'phone number',
        textAlign: TextAlign.left,
        keyboardType:
TextInputType.number,
    ),
    ),
  ],
),
),
const SizedBox(height: 20),
Text(
  'Carrier charges may apply',
  style: TextStyle(
    color: context.theme.greyColor,
  ),

```

```

    ),
  ],
),
floatingActionButtonLocation:
FloatingActionButtonLocation.centerFloat,
floatingActionButton:
CustomElevatedButton(
  onPressed: sendCodeToPhone,
  text: 'NEXT',
  buttonWidth: 90,
),
);
}
}

```

UserInfo Page:

```

import 'dart:io';
import 'dart:typed_data';

import 'package:flutter/material.dart';
import
'package:flutter_riverpod/flutter_riverpod.da
rt';
import
'package:image_picker/image_picker.dart';
import
'package:whatsapp_messenger/common/exte
nsion/custom_theme_extension.dart';
import
'package:whatsapp_messenger/common/hel
per/show_alert_dialog.dart';
import
'package:whatsapp_messenger/common/util
s/colors.dart';
import
'package:whatsapp_messenger/common/wid
gets/custom_elevated_button.dart';
import
'package:whatsapp_messenger/common/wid
gets/custom_icon_button.dart';

```

```

import
'package:whatsapp_messenger/common/wid
gets/short_h_bar.dart';
import
'package:whatsapp_messenger/feature/auth/
controller/auth_controller.dart';
import
'package:whatsapp_messenger/feature/auth/
pages/image_picker_page.dart';
import
'package:whatsapp_messenger/feature/auth/
widgets/custom_text_field.dart';

class UserInfoPage extends
ConsumerStatefulWidget {
  const UserInfoPage({super.key,
this.profileImageUrl});

  final String? profileImageUrl;

  @override
  ConsumerState<UserInfoPage>
createState() => _UserInfoPageState();
}

```

```

class _UserInfoPageState extends
ConsumerState<UserInfoPage> {
  File? imageCamera;
  Uint8List? imageGallery;
  late TextEditingController
usernameController;

  @override
  void initState() {
    super.initState();
    usernameController =
TextEditingController();
  }

  @override
  void dispose() {
    usernameController.dispose();
    super.dispose();
  }

  Future<void> saveUserDataToFirebase()
async {
    String username =
usernameController.text.trim();

    if (username.isEmpty) {
      showAlertDialog(context: context,
message: 'Please provide a username');
      return;
    }
    if (username.length < 3 || username.length
> 20) {
      showAlertDialog(
        context: context, message: 'Username
length should be between 3-20');
      return;
    }

    await

```

```

ref.read(authControllerProvider).saveUserIn
foToFirestore(
  username: username,
  profileImage:
    imageCamera ?? imageGallery ??
widget.profileImageUrl ?? "",
  context: context,
);
}

void imagePickerTypeBottomSheet() {
  showModalBottomSheet(
    context: context,
    builder: (context) {
      return Padding(
        padding: const
EdgeInsets.symmetric(vertical: 10),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.min,
          children: [
            const ShortHBar(),
            Padding(
              padding: const
EdgeInsets.symmetric(horizontal: 20),
              child: Row(
                children: [
                  const Text(
                    'Profile photo',
                    style:
                      TextStyle(fontSize: 20,
fontWeight: FontWeight.w500),
                  ),
                  const Spacer(),
                  CustomIconButton(
                    onPressed: () =>
Navigator.pop(context),
                    icon: Icons.close,
                  ),
                ],
              ),
            ),
          ],
        ),
      ),
    ),
  ),
)

```

```

    ),
    Divider(color:
context.theme.greyColor!.withOpacity(.3)),
    Row(
      mainAxisAlignment:
MainAxisAlignment.spaceEvenly,
      children: [
        imagePickerIcon(
          onTap: pickImageFromCamera,
          icon:
Icons.camera_alt_rounded,
          text: 'Camera',
        ),
        imagePickerIcon(
          onTap: pickImageFromGallery,
          icon:
Icons.photo_camera_back_rounded,
          text: 'Gallery',
        ),
      ],
    ),
  ],
),
);
},
);
}

```

```

Future<void> pickImageFromCamera()
async {
  Navigator.pop(context);
  final image = await
ImagePicker().pickImage(source:
ImageSource.camera);
  if (image != null) {
    setState() {
      imageCamera = File(image.path);
      imageGallery = null;
    };
  }
}

```

```

Future<void> pickImageFromGallery()
async {
  Navigator.pop(context);
  final image = await Navigator.push(
    context,
    MaterialPageRoute(builder: (context) =>
const ImagePickerPage()),
  );
  if (image != null) {
    setState() {
      imageGallery = image;
      imageCamera = null;
    };
  }
}

```

```

Widget imagePickerIcon({
  required VoidCallback onTap,
  required IconData icon,
  required String text,
}) {
  return Column(
    children: [
      CustomIconButton(
        onPressed: onTap,
        icon: icon,
        iconColor: Colors.greenDark,
        minWidth: 50,
        border: Border.all(color:
context.theme.greyColor!.withOpacity(.2)),
      ),
      const SizedBox(height: 5),
      Text(text, style: TextStyle(color:
context.theme.greyColor)),
    ],
  );
}

```

```

@override
Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      backgroundColor:
Theme.of(context).colorScheme.surface,
      elevation: 0,
      title: Text(
        'Profile info',
        style: TextStyle(color:
context.theme.authAppBarTextColor),
      ),
      centerTitle: true,
    ),
    body: SingleChildScrollView(
      padding: const
EdgeInsets.symmetric(horizontal: 20),
      child: Column(
        children: [
          Text(
            'Please provide your name and an
optional profile photo',
            textAlign: TextAlign.center,
            style: TextStyle(color:
context.theme.greyColor),
          ),
          const SizedBox(height: 40),
          GestureDetector(
            onTap:
imagePickerTypeBottomSheet,
            child: Container(
              padding: const EdgeInsets.all(26),
              decoration: BoxDecoration(
                shape: BoxShape.circle,
                color:
context.theme.photoIconBgColor,
                border: Border.all(
                  color: (imageCamera == null
&&
                    imageGallery == null &&
                    widget.profileImageUrl ==

```

```

null)
                ? Colors.transparent
                :
context.theme.greyColor!.withOpacity(.4),
            ),
            image: (imageCamera != null ||
                imageGallery != null ||
                widget.profileImageUrl !=
null)
                ? DecorationImage(
                    fit: BoxFit.cover,
                    image: imageGallery != null
?
MemoryImage(imageGallery!)
: widget.profileImageUrl
!= null
?
NetworkImage(widget.profileImageUrl!)
:
FileImage(imageCamera!) as
ImageProvider,
                )
                : null,
            ),
            child: Icon(
              Icons.add_a_photo_rounded,
              size: 48,
              color: (imageCamera == null &&
                imageGallery == null &&
                widget.profileImageUrl ==
null)
                ?
context.theme.photoIconColor
                : Colors.transparent,
            ),
            ),
            ),
            const SizedBox(height: 40),
            CustomTextField(
              controller: usernameController,
              hintText: 'Type your name here',

```

```

        autoFocus: true),
    ],
  ),
),
floatingActionButton:
CustomElevatedButton(
  onPressed: saveUserDataToFirestore,

```

```

text: 'NEXT', buttonWidth: 90),
  );
}
}

```

Verification Page:

```

import 'package:flutter/material.dart';
import
'package:flutter_riverpod/flutter_riverpod.dart';
import
'package:whatsapp_messenger/common/extension/custom_theme_extension.dart';
import
'package:whatsapp_messenger/feature/auth/controller/auth_controller.dart';
import
'package:whatsapp_messenger/feature/auth/widgets/custom_text_field.dart';

class VerificationPage extends
ConsumerWidget {
  const VerificationPage({
    super.key,
    required this.smsCodeId,
    required this.phoneNumber,
  });

  final String smsCodeId;
  final String phoneNumber;

  Future<void> verifySmsCode(
    BuildContext context,
    WidgetRef ref,
    String smsCode,
  ) async {
    await
    ref.read(authControllerProvider).verifySmsCode
    (
      context: context,
      smsCodeId: smsCodeId,

```

```

      smsCode: smsCode,
    );
  }

  @override
  Widget build(BuildContext context,
  WidgetRef ref) {
    return Scaffold(
      appBar: AppBar(
        backgroundColor:
Theme.of(context).scaffoldBackgroundColor,
        elevation: 0,
        centerTitle: true,
        title: Text(
          'Verify your number',
          style: TextStyle(
            color:
context.theme.authAppBarTextColor,
          ),
        ),
        actions: [
          IconButton(
            onPressed: () {},
            icon: const Icon(Icons.more_vert),
          ),
        ],
      ),
      body: SingleChildScrollView(
        padding: const
EdgeInsets.symmetric(horizontal: 20),
        child: Column(
          children: [
            Padding(
              padding: const

```

```

EdgeInsets.symmetric(horizontal: 10),
  child: RichText(
    textAlign: TextAlign.center,
    text: TextSpan(
      style: TextStyle(color:
context.theme.greyColor),
    children: [
      const TextSpan(
        text:
          "You've tried to register before
requesting an SMS or Call with your code.",
      ),
      TextSpan(
        text: " Wrong number?",
        style: TextStyle(
          color: context.theme.blueColor,
        ),
      ),
    ],
  ),
),
const SizedBox(height: 20),
Container(
  padding: const
EdgeInsets.symmetric(horizontal: 80),
  child: CustomTextField(
    hintText: "- - - - -",
    fontSize: 30,
    autoFocus: true,
    keyboardType:
TextInputType.number,
    onChanged: (value) async {
      if (value.length == 6) {
        await verifySmsCode(context, ref,
value);
      }
    },
  ),
),
const SizedBox(height: 20),
Text(
  'Enter 6-digit code',

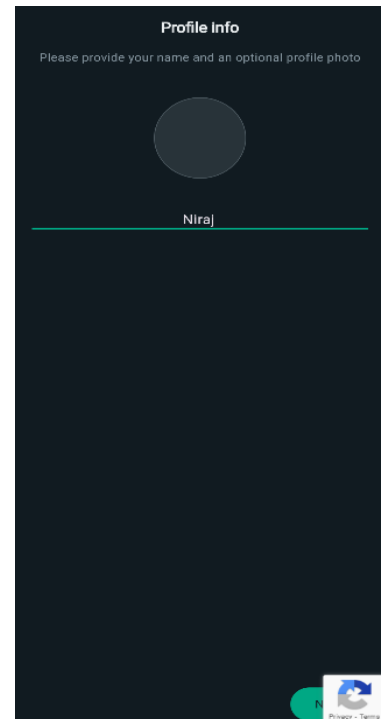
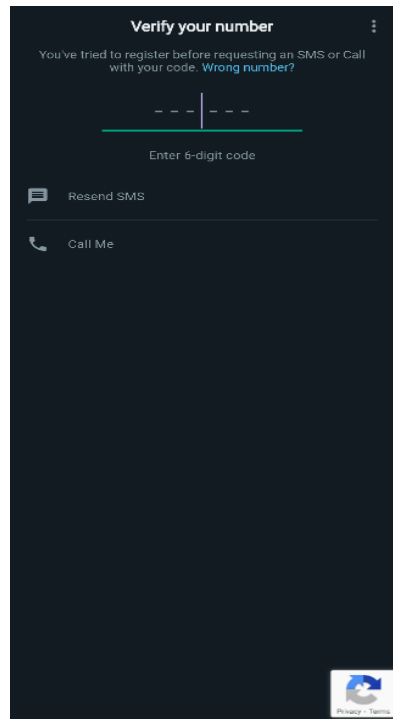
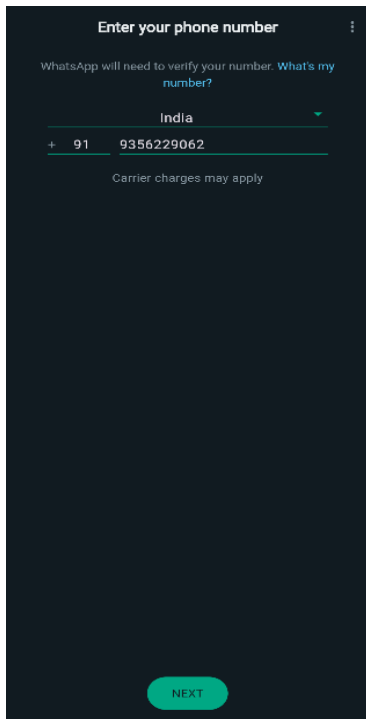
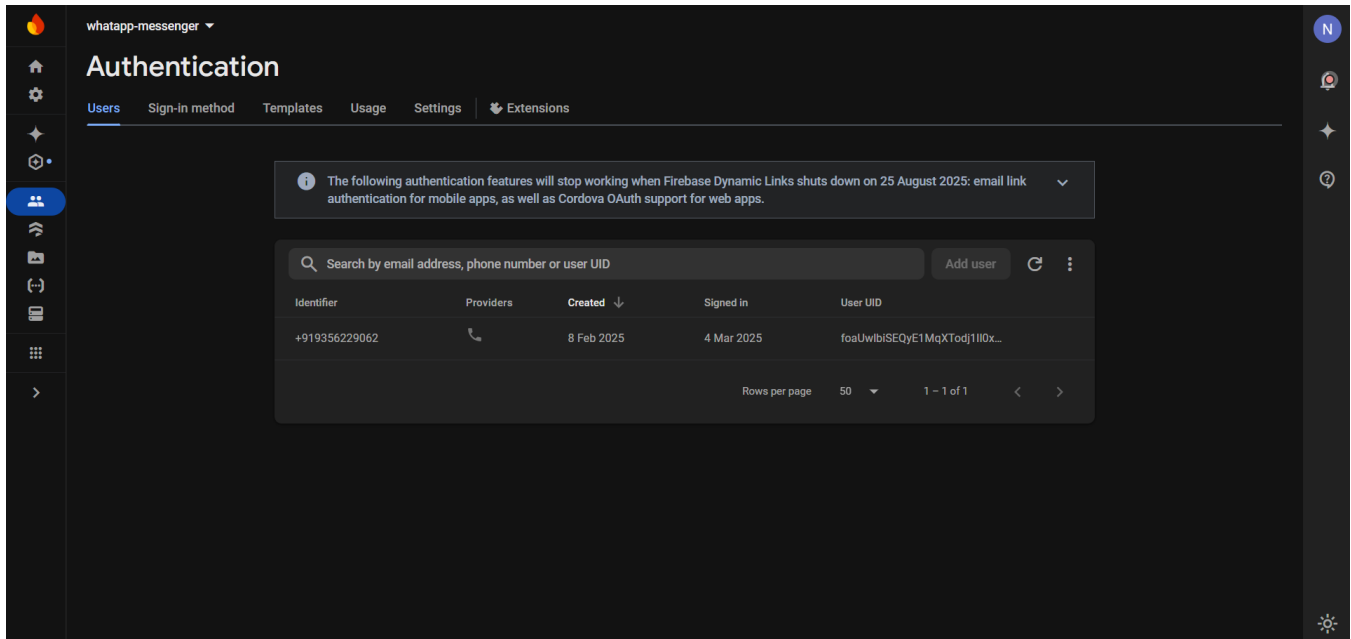
```

```

    style: TextStyle(color:
context.theme.greyColor),
  ),
  const SizedBox(height: 30),
  Row(
    children: [
      Icon(Icons.message, color:
context.theme.greyColor),
      const SizedBox(width: 20),
      Text(
        'Resend SMS',
        style: TextStyle(
          color: context.theme.greyColor,
        ),
      ),
    ],
  ),
  const SizedBox(height: 10),
  Divider(
    color:
context.theme.greyColor!.withOpacity(.2),
  ),
  const SizedBox(height: 10),
  Row(
    children: [
      Icon(Icons.phone, color:
context.theme.greyColor),
      const SizedBox(width: 20),
      Text(
        'Call Me',
        style: TextStyle(
          color: context.theme.greyColor,
        ),
      ),
    ],
  ),
),
);
}
}

```

After Registering, the user details is saved in the database.



User details get fetched from the database in the profile page.

The screenshot shows the Google Cloud Firestore console for a project named 'whatapp-messenger'. The 'Data' tab is selected, and the breadcrumb navigation shows the path: 'users' > 'foaUwlbISEQyE1MqXTodj1lI0xm1'. The document is expanded, showing its fields:

- active: true
- groupId
- lastSeen: 1741102062460
- phoneNumber: "+919356229062"
- profileImageUrl: ""
- uid: "foaUwlbISEQyE1MqXTodj1lI0xm1"
- username: "ABC"

The URL at the bottom of the console is: https://console.firebase.google.com/u/0/project/whatapp-messenger-8275a/firestore?fb_gclid=CjwKCAIA5pq-BhBuEiwAvkzVZctzmR79BRs232e4HoELVSN04ihCycmdUkn0f9hih7OW3Pfr-ECuRoChAUQAvD_BwE